Comparison of abmneo and ABM - ABM part

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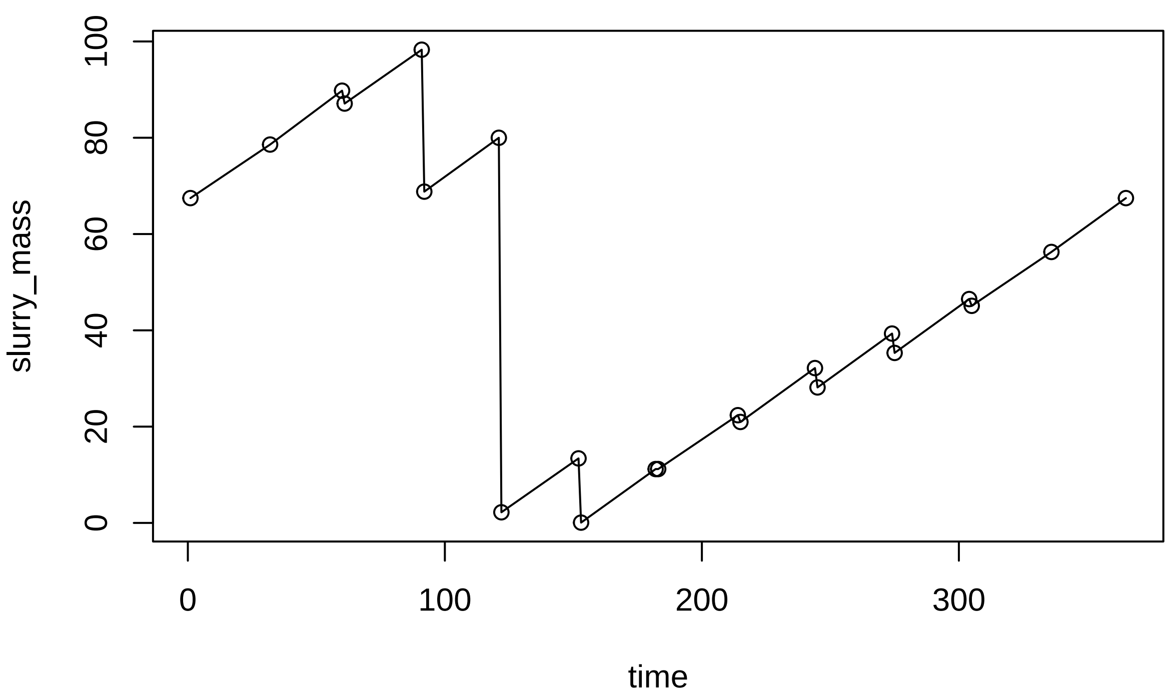
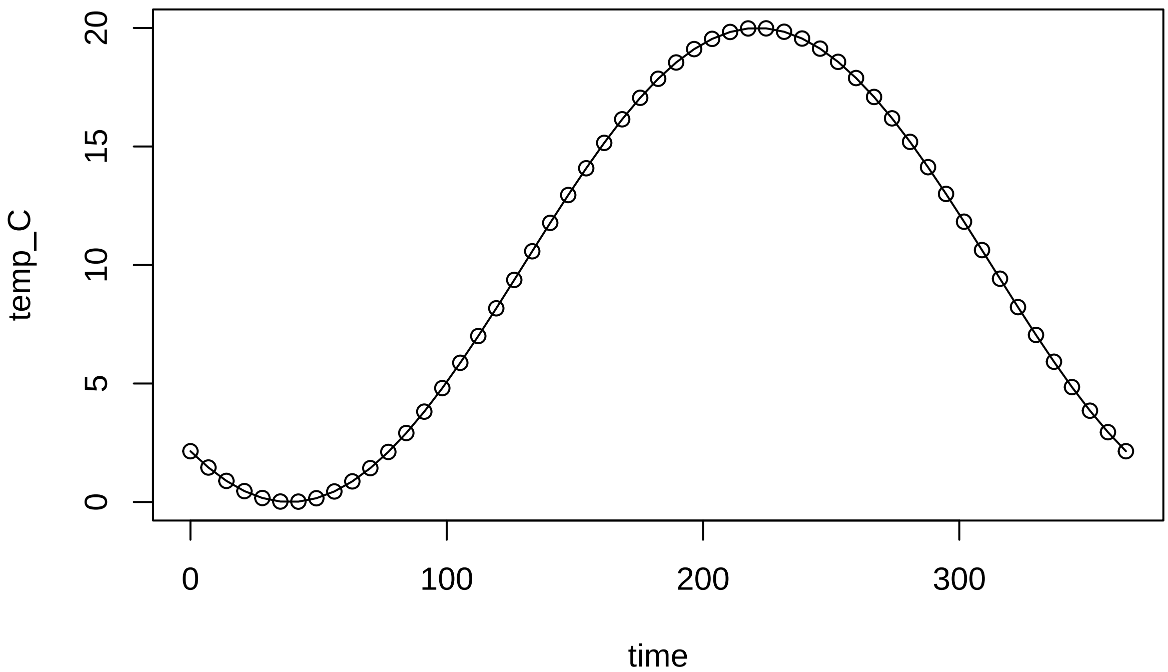
```
devtools::load_all('../../ABM')
```

```
## i Loading ABM
```

```
mass_series <- read.csv('level_pig_od_ref.csv')
plot(slurry_mass ~ time, data = mass_series, type = 'o')
```




```
temp_dat <- data.frame(time = seq(0, 365, length.out = 53))
temp_dat$temp_C <- 10 + 10 * sin((temp_dat$time - 130) / 365 * 2 * pi)
plot(temp_C ~ time, data = temp_dat, type = 'o')
```




```
arrh_pars <- list(lnA =  29.96298,
              E_CH4 = 81000,
              A = c(xa_dead= 3.61383 * 10^12, starch = 5 * 3.61383 * 10^12, Cfat = 0.1 * 3.61383 * 10^1
              E = c(xa_dead= 81557, starch = 81557, Cfat = 81557, CPs = 81557, CPf = 81557, RFd = 81557
              R = 8.314,
              VS_CH4 = 6.67,
              scale_alpha_opt = list(VSd = 0.46056, notVSd = 0.556886, CP = 2.065),
              kl = c(NH3 = 336, NH3_floor = 0.36*337, H2S = 0.02),
              scale_EF_NH3 = 0.0000048)
```

```
grp_pars <- list(grps = c('m0', 'm1', 'm2','m3','m4','m5','sr1'),
                 yield = c(default = 0.05, sr1 = 0.065),
                 xa_fresh = c(m0 = 0.06, m1 = 0.06, m2 = 0.06, m3 = 0.06, m4 = 0.06, m5 = 0.06, sr1 = 0
                 xa_init = c(all = 0.1),
                 decay_rate = c(all = 0.02),
                 ks_coefficient = c(default = 1.153337, sr1 = 0.461335),
                 qhat_opt =  c(m0 = 0.46289, m1 = 0.74568, m2 = 0.9006605, m3 = 2.79672, m4 = 4.66012, 
                 T_opt = c(m0 = 18, m1 = 18, m2 = 28, m3 = 36, m4 = 43.75, m5 = 55, sr1 = 43.75),
                 T_min = c(m0 = 0, m1 = 9.67, m2 = 9.67, m3 = 15, m4 = 26.25, m5 = 30, sr1 = 0),
                 T_max = c(m0 = 25, m1 = 25, m2 = 38, m3 = 45, m4 = 51.25, m5 = 60, sr1 = 51.25),
                 ki_NH3_min = c(all = 0.015),
                 ki_NH3_max = c(all = 5.37),
                 ki_NH4_min = c(all = 2.7),
                 ki_NH4_max = c(all = 6.8492),
                 ki_H2S_slope = c(default = -0.10623, sr1 = -0.1495),
                 ki_H2S_int = c(default = 1.02, sr1 = 1.42),
```

```
                 ki_H2S_min = c(default = 0.08),
                 IC50_low = c(default = 0.20854, sr1 = 0.2772),
                 ki_HAC = c(default = 0.31),
                 pH_UL = c(default = 100),
                 pH_LL = c(default = -100))
```

Write mng_pars in order to insert slurry_mass data

```
mng_pars <- list(slurry_prod_rate = 5700,
                slurry_mass = mass_series,
                storage_depth = 2,
                resid_depth = 0,
                floor_area = 0,
                area = 0.03,
                empty_int = 0,
                temp_C = temp_dat,
                wash_water = 0,
                wash_int = NA,
                rest_d = 0,
                cover = 'none',
                resid_enrich = 0.9,
                graze = c(start = 'May', duration = 0, hours_day = 0),
                scale = c(ks_coefficient = 1, qhat_opt = 1, xa_fresh = 1, yield = 1, alpha_opt = 1)
)
```

Take man_pars from man_pars_pig2.0

```
man_pars_pig2.0 <- list(conc_fresh = list(sulfide = 0.01, urea = 3.17, sulfate = 0.01, TAN = 0.0, starc
                                    VFA = 1.7, xa_dead = 0, Cfat = 27.6, CPs = 21.1 * 0.55, CPf = 21.
                                    ash = 15), pH = 7, dens = 1000)
```

```
devtools::load_all('../../ABM')
```

```
## i Loading ABM
```

```
system.time(
  abm_out <- abm(
    days = 365,
    man_pars = man_pars_pig2.0,
    mng_pars = mng_pars,
    arrh_pars = arrh_pars,
    grp_pars = grp_pars,
    add_pars = list(evap = 0, rain = 0, COD_conv.CH4 = 5.32),
    resp = FALSE,
    pH_inhib_overrule = TRUE,
    startup = 2
  )
)
```

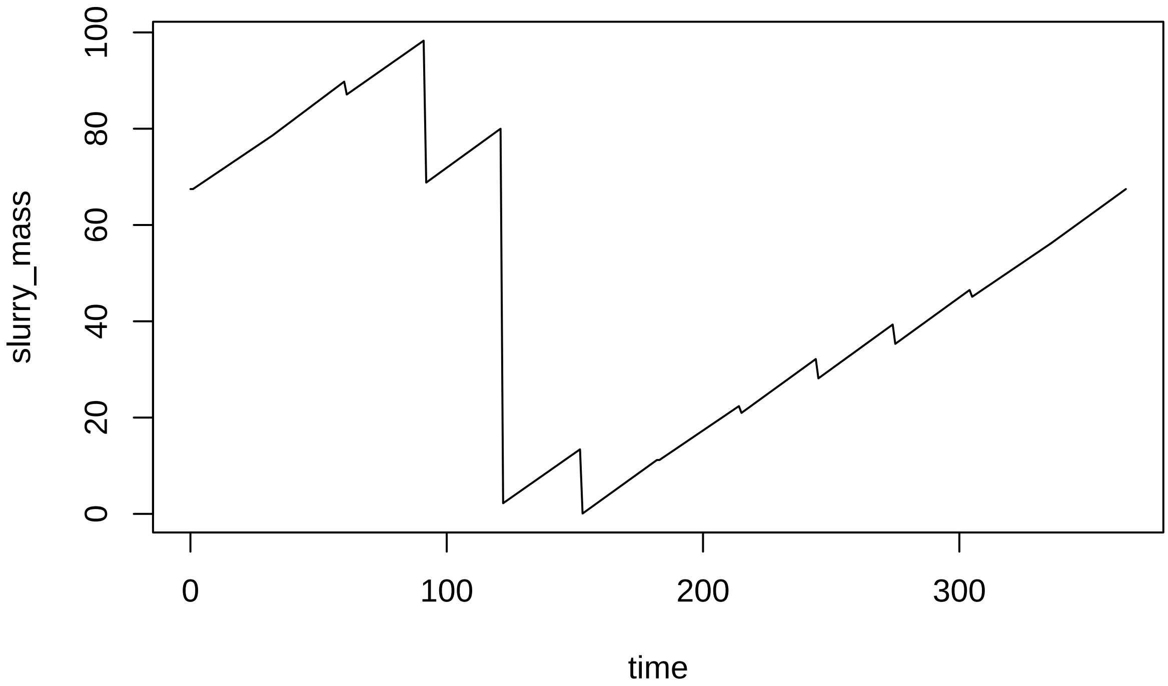
```
##
## Startup run 1x ->
```

```
## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurr
## Check output.
```

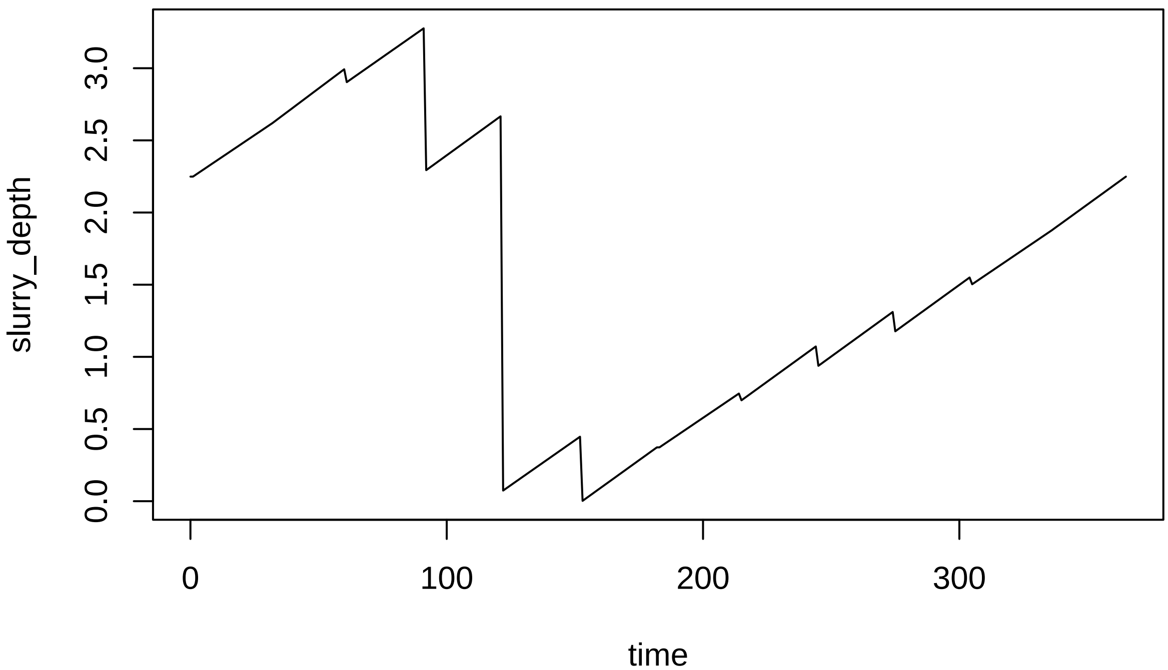
```
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
```

```
## evaporation = 0 kg/m2/day

## 2x ->

## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
## Check output.

## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day

## and final run

## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
## Check output.

## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day

##    user  system elapsed
##   1.992   0.000   2.004
```

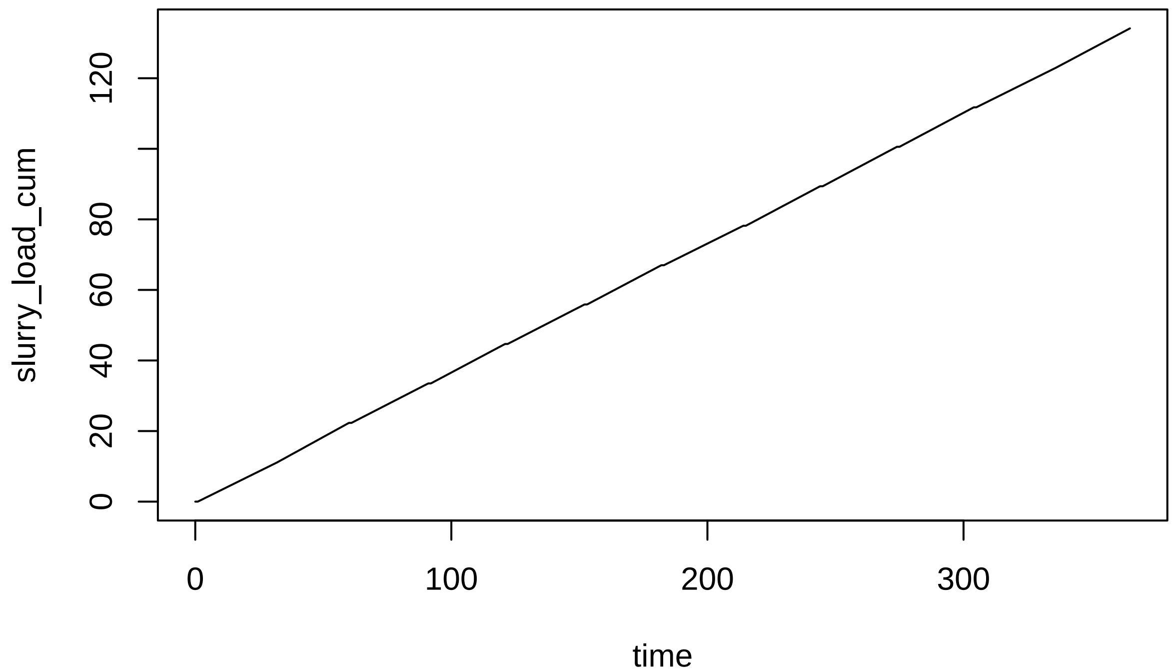
```
plot(slurry_mass ~ time, data = abm_out, type = 'l')
```



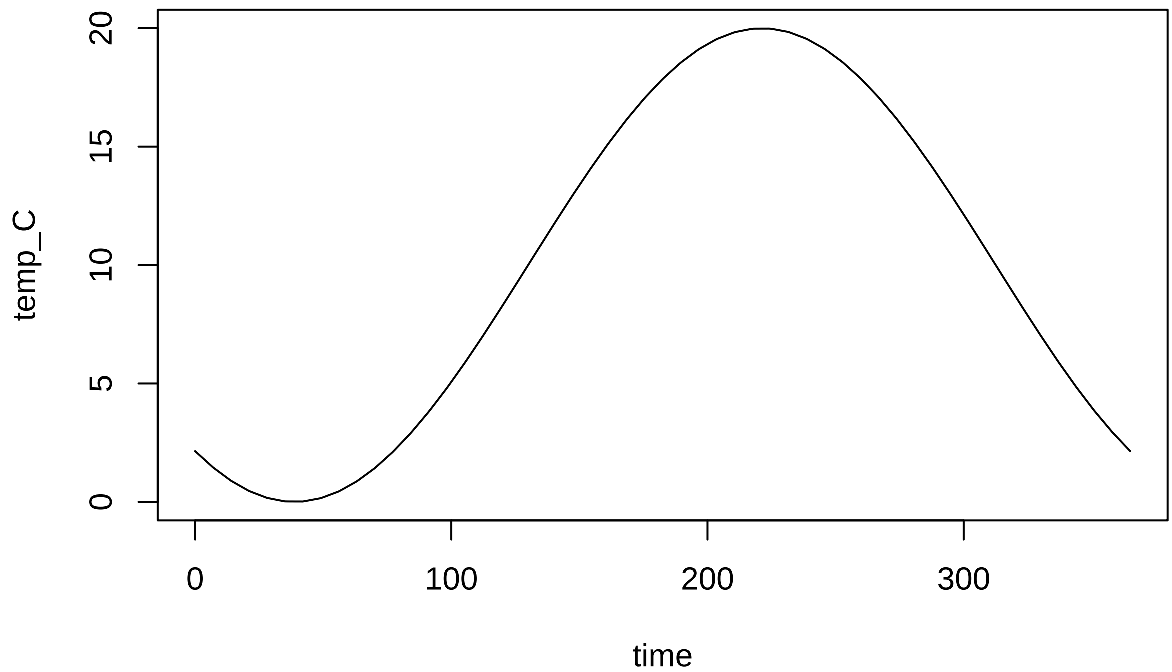

```
plot(slurry_depth ~ time, data = abm_out, type = 'l')
```



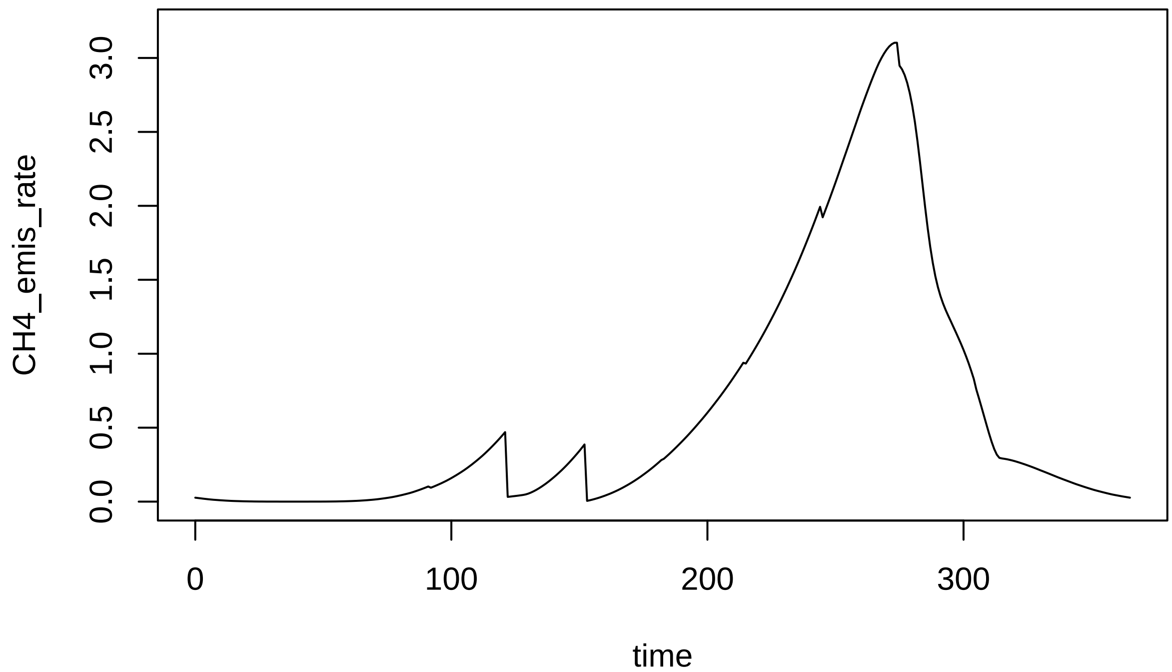

```r
plot(slurry_load_cum ~ time, data = abm_out, type = 'l')
```



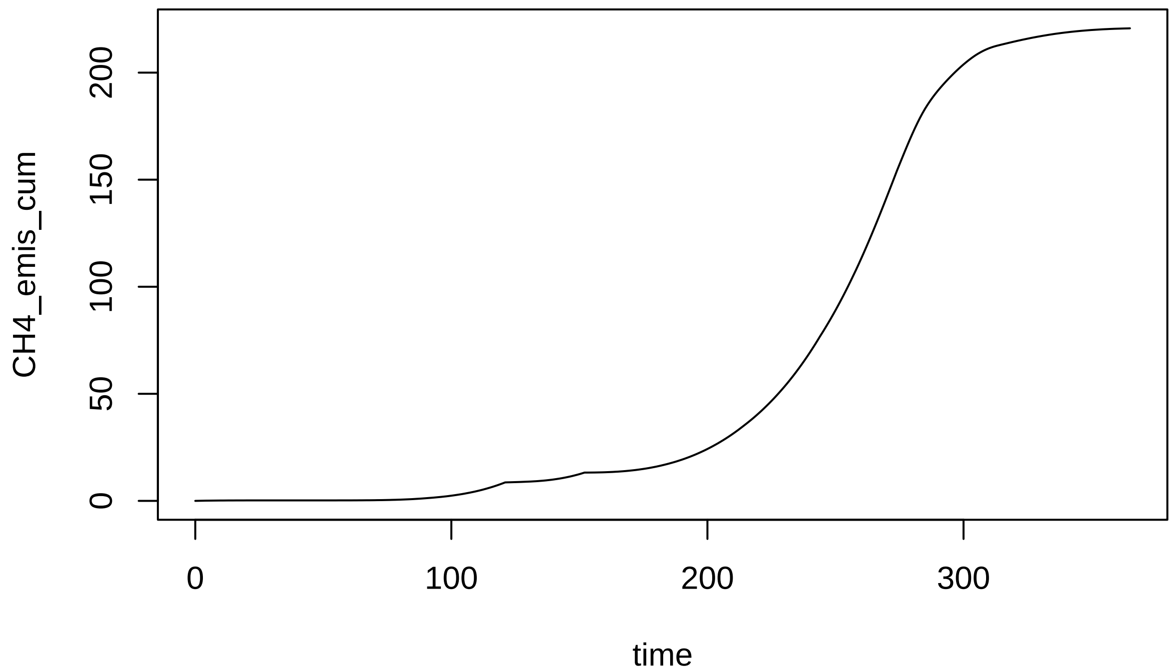

```
plot(temp_C ~ time, data = abm_out, type = 'l')
```



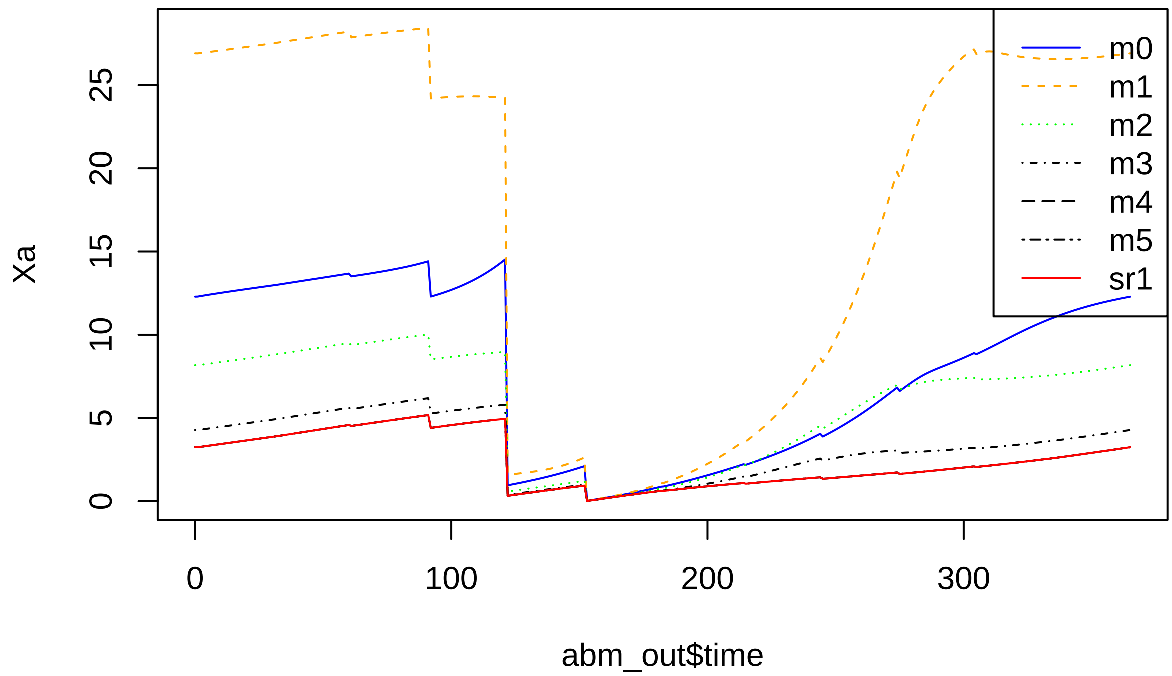

```
plot(CH4_emis_rate ~ time, data = abm_out, type = 'l', ylim = c(0, 3.2))
```



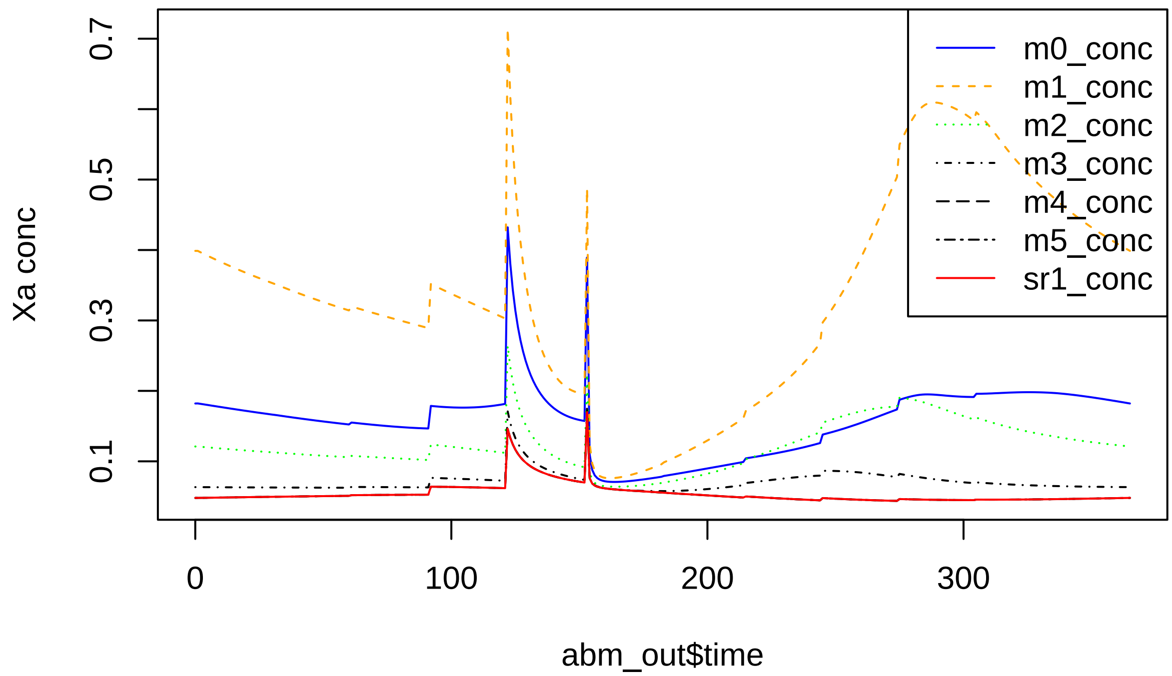

```r
plot(CH4_emis_cum ~ time, data = abm_out, type = 'l')
```



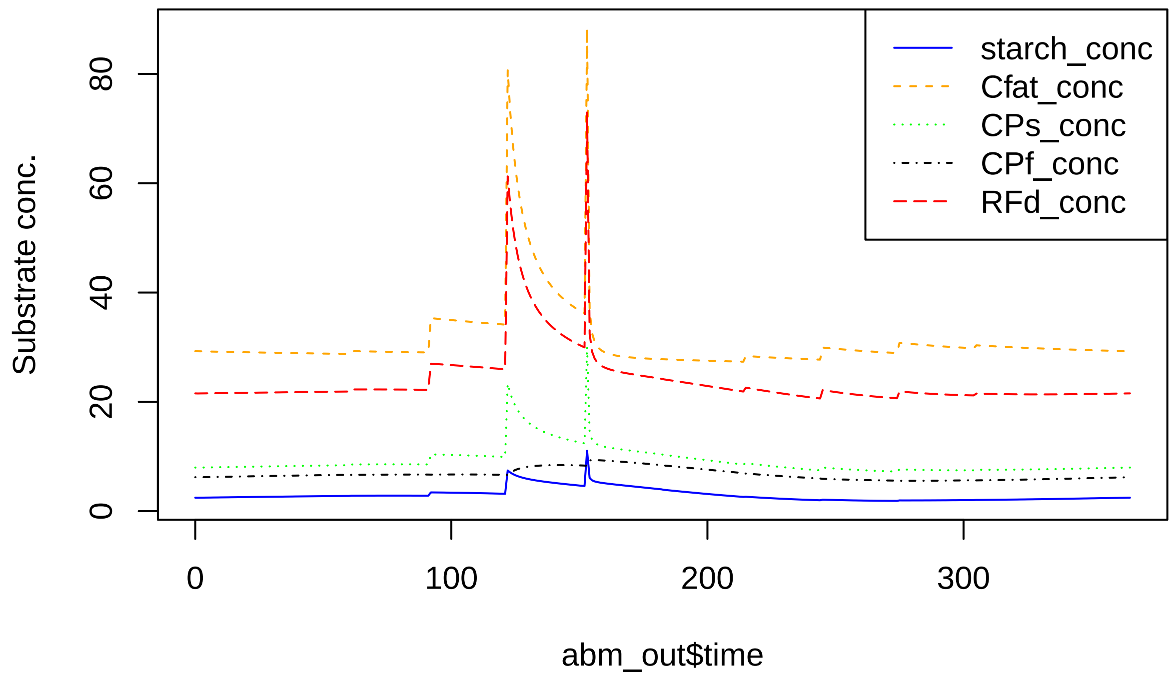

```r
matplot(
  abm_out$time,
  abm_out[, nn <- grp_pars$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



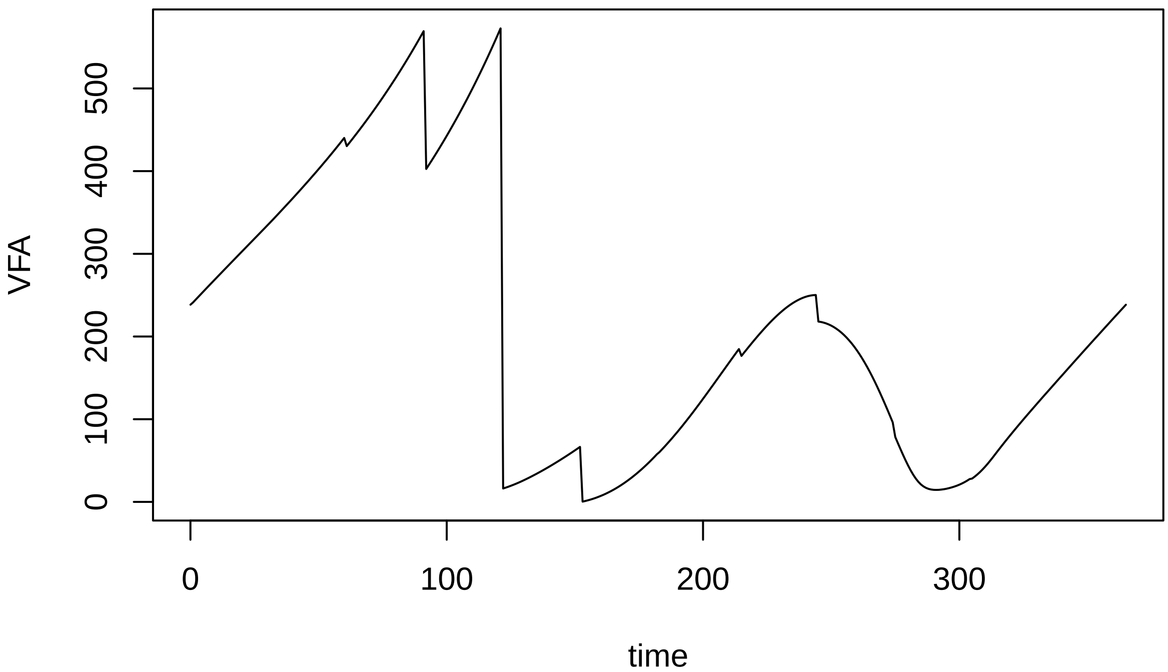

```r
matplot(
  abm_out$time,
  abm_out[, nn <- paste0(grp_pars$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



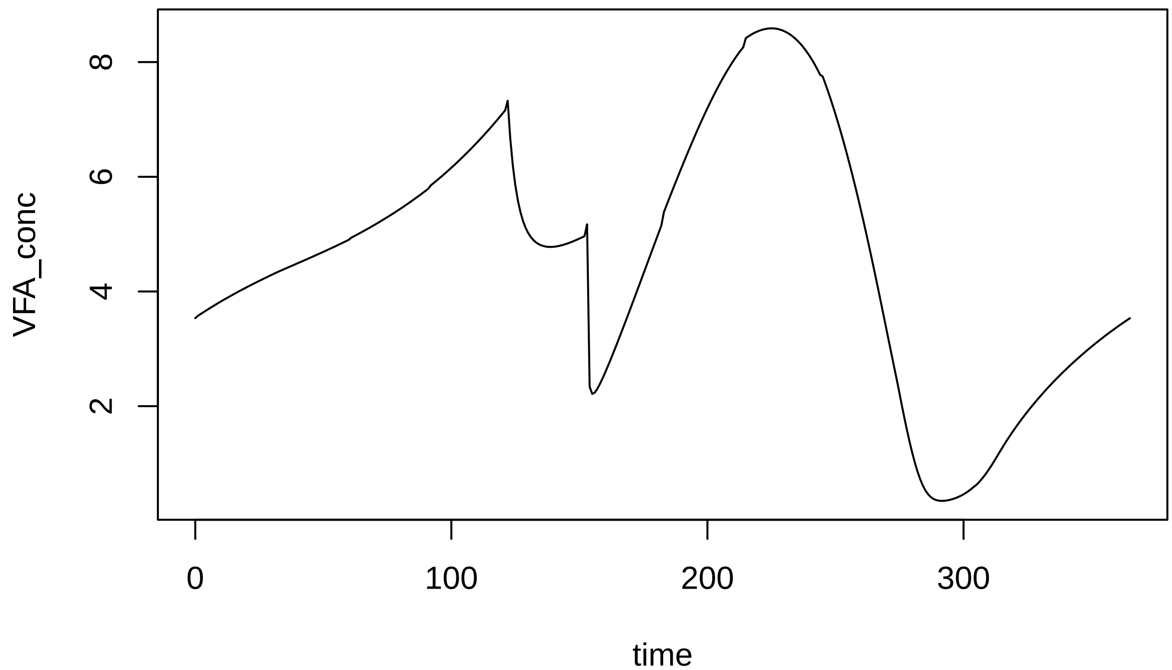

```r
matplot(
  abm_out$time,
  abm_out[, nn <- paste0(c('starch', 'Cfat', 'CPs', 'CPf', 'RFd'), '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```




```
plot(VFA ~ time, data = abm_out, type = 'l')
```




```r
plot(VFA_conc ~ time, data = abm_out, type = 'l')
```



No methanogens, for comparison

```
grp_pars0 <- list(grps = c('m0', 'm1', 'm2','m3','m4','m5','sr1'),
                  yield = c(default = 0.05, sr1 = 0.065),
                  xa_fresh = c(all = 0),
                  xa_init = c(all = 0),
                  decay_rate = c(all = 0.02),
                  ks_coefficient = c(default = 1.153337, sr1 = 0.461335),
                  qhat_opt =  c(m0 = 0.46289, m1 = 0.74568, m2 = 0.9006605, m3 = 2.79672, m4 = 4.66012, 
                  T_opt = c(m0 = 18, m1 = 18, m2 = 28, m3 = 36, m4 = 43.75, m5 = 55, sr1 = 43.75),
                  T_min = c(m0 = 0, m1 = 9.67, m2 = 9.67, m3 = 15, m4 = 26.25, m5 = 30, sr1 = 0),
                  T_max = c(m0 = 25, m1 = 25, m2 = 38, m3 = 45, m4 = 51.25, m5 = 60, sr1 = 51.25),
                  ki_NH3_min = c(all = 0.015),
                  ki_NH3_max = c(all = 5.37),
                  ki_NH4_min = c(all = 2.7),
                  ki_NH4_max = c(all = 6.8492),
                  ki_H2S_slope = c(default = -0.10623, sr1 = -0.1495),
                  ki_H2S_int = c(default = 1.02, sr1 = 1.42),
                  ki_H2S_min = c(default = 0.08),
                  IC50_low = c(default = 0.20854, sr1 = 0.2772),
                  ki_HAC = c(default = 0.31),
                  pH_UL = c(default = 100),
                  pH_LL = c(default = -100))
```

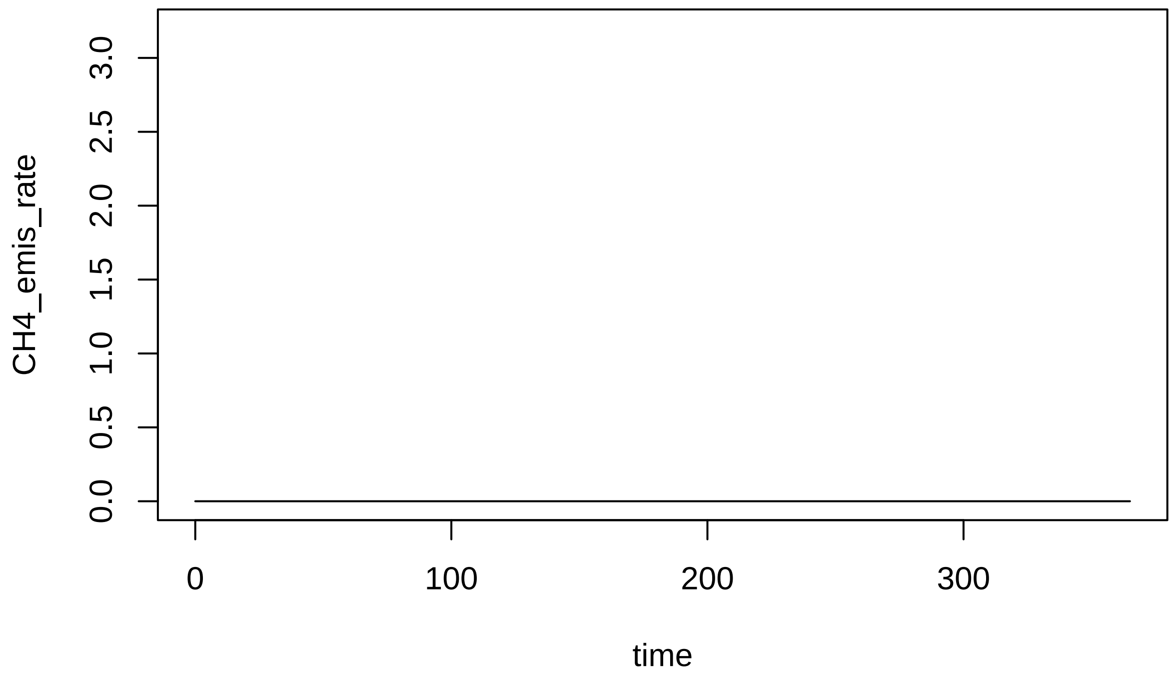
```r
devtools::load_all('../../ABM')
```

```
## i Loading ABM
```

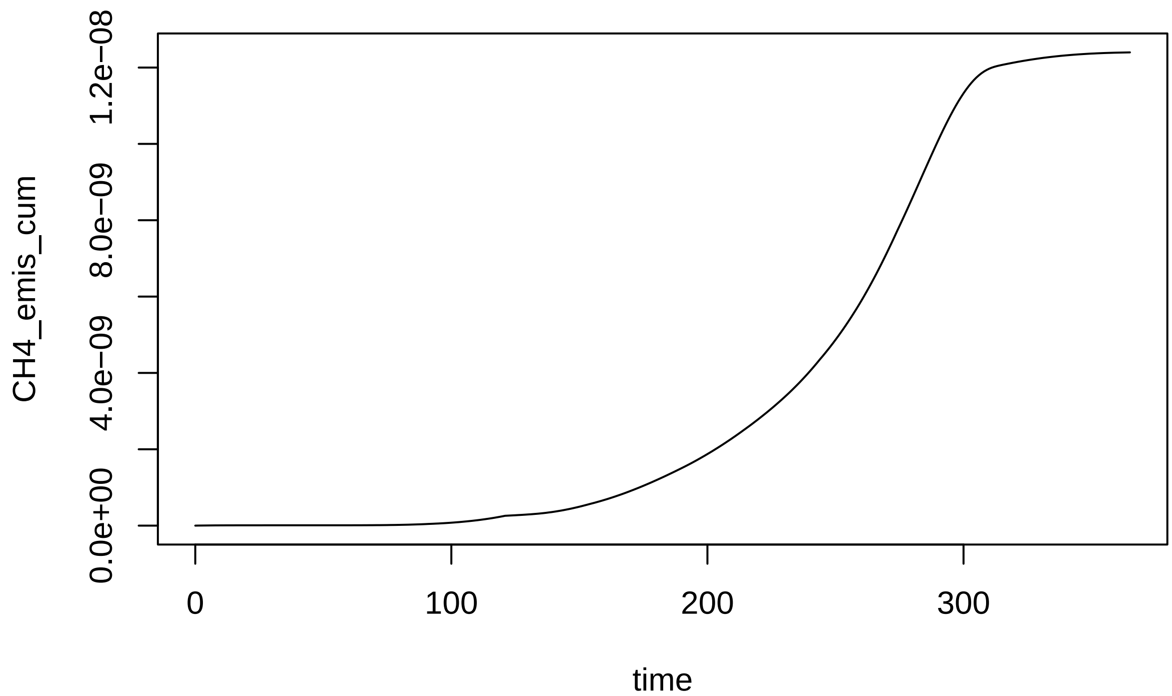
```r
system.time(
  abm_out0 <- abm(
    days = 365,
    man_pars = man_pars_pig2.0,
    mng_pars = mng_pars,
    arrh_pars = arrh_pars,
    grp_pars = grp_pars0,
    add_pars = list(evap = 0, rain = 0, COD_conv.CH4 = 5.32),
    resp = FALSE,
    pH_inhib_overrule = TRUE,
    startup = 2
  )
)
```

```
##
## Startup run 1x ->

## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurr
## Check output.

## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day

## 2x ->

## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurr
## Check output.

## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day

## and final run

## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurr
## Check output.

## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day

##    user  system elapsed
##   2.505   0.000   2.505
```

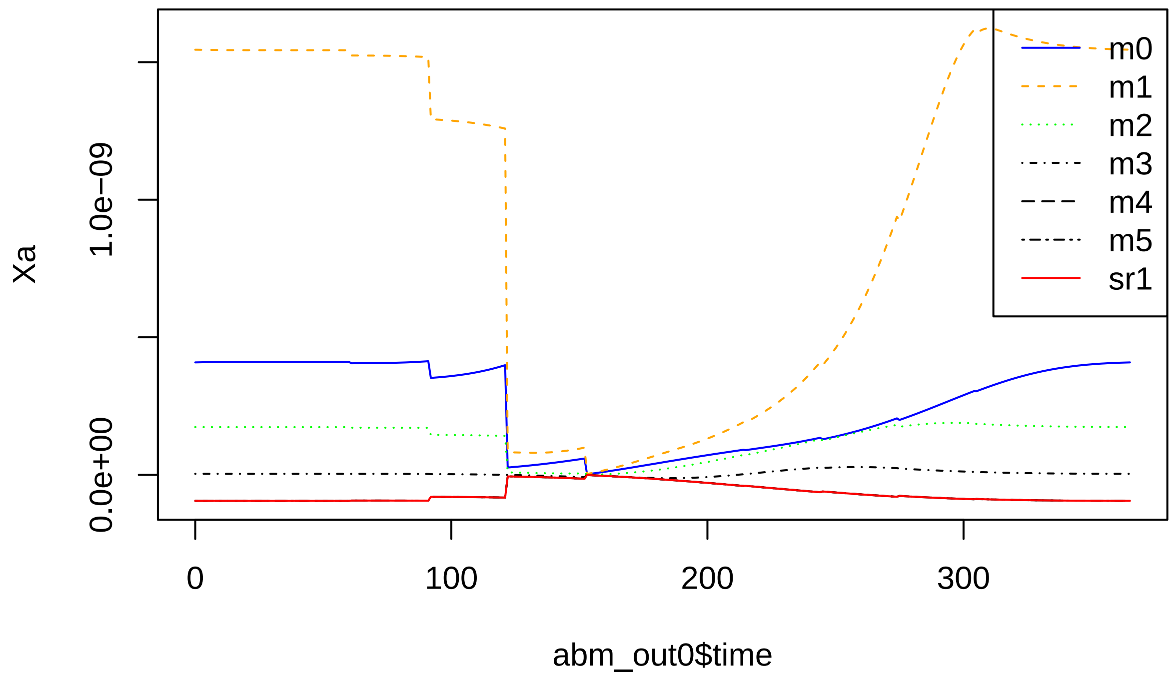
```r
plot(CH4_emis_rate ~ time, data = abm_out0, type = 'l', ylim = c(0, 3.2))
```



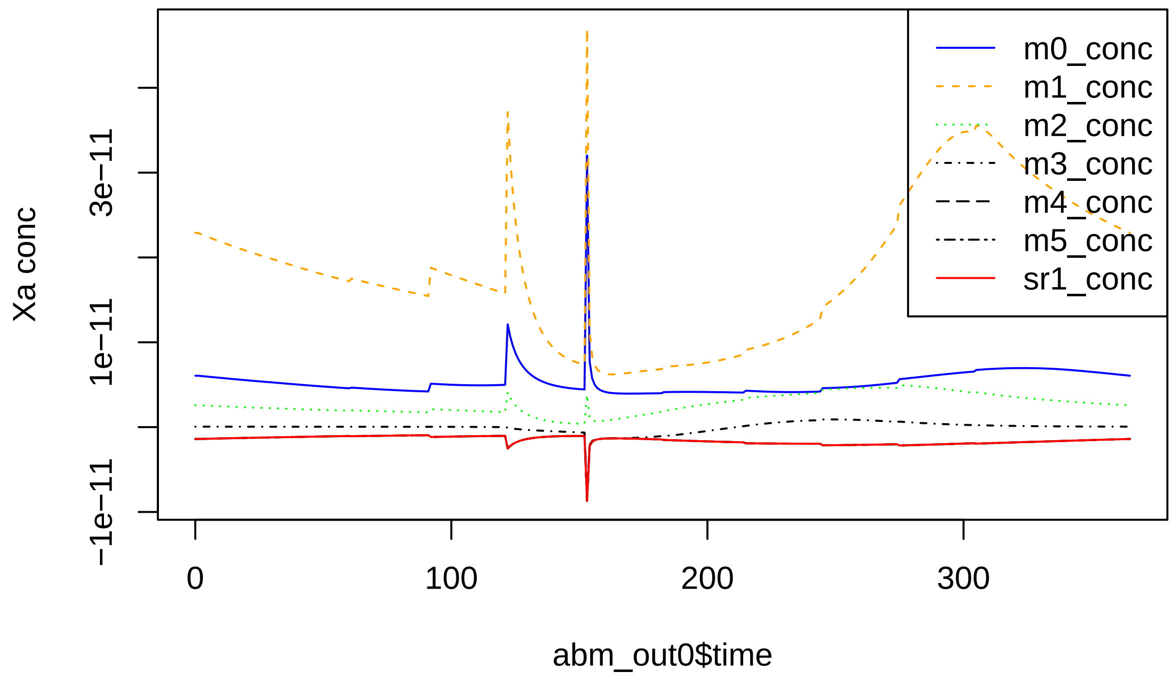

```r
plot(CH4_emis_cum ~ time, data = abm_out0, type = 'l')
```



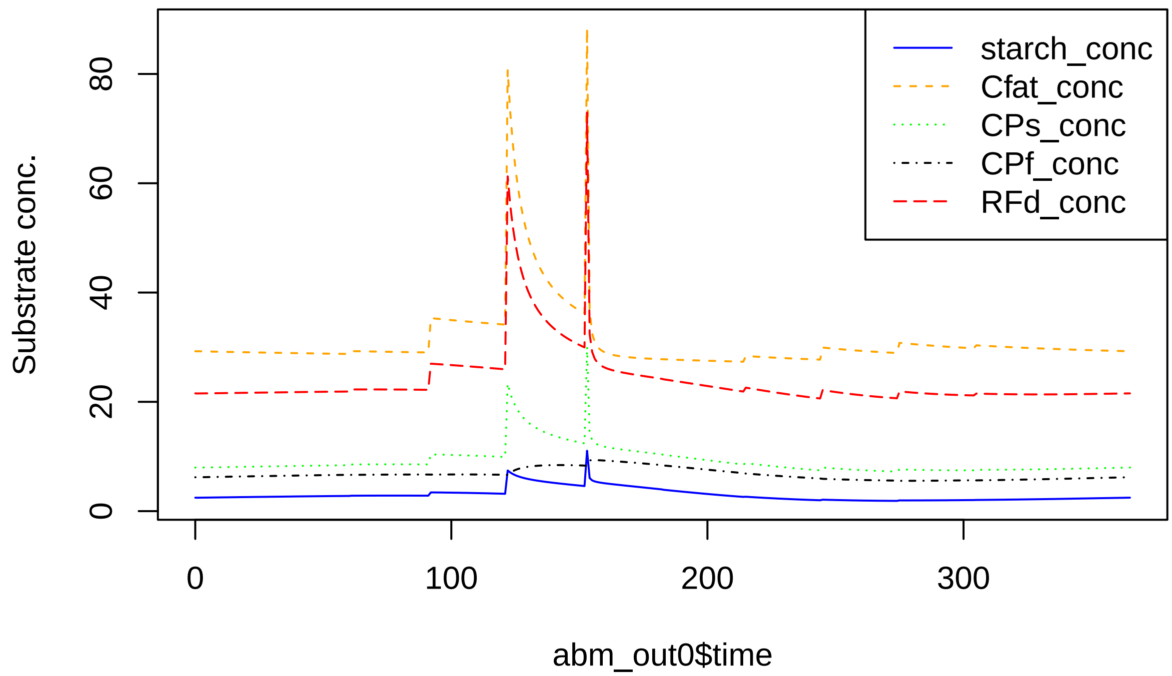

```
matplot(
  abm_out0$time,
  abm_out0[, nn <- grp_pars$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



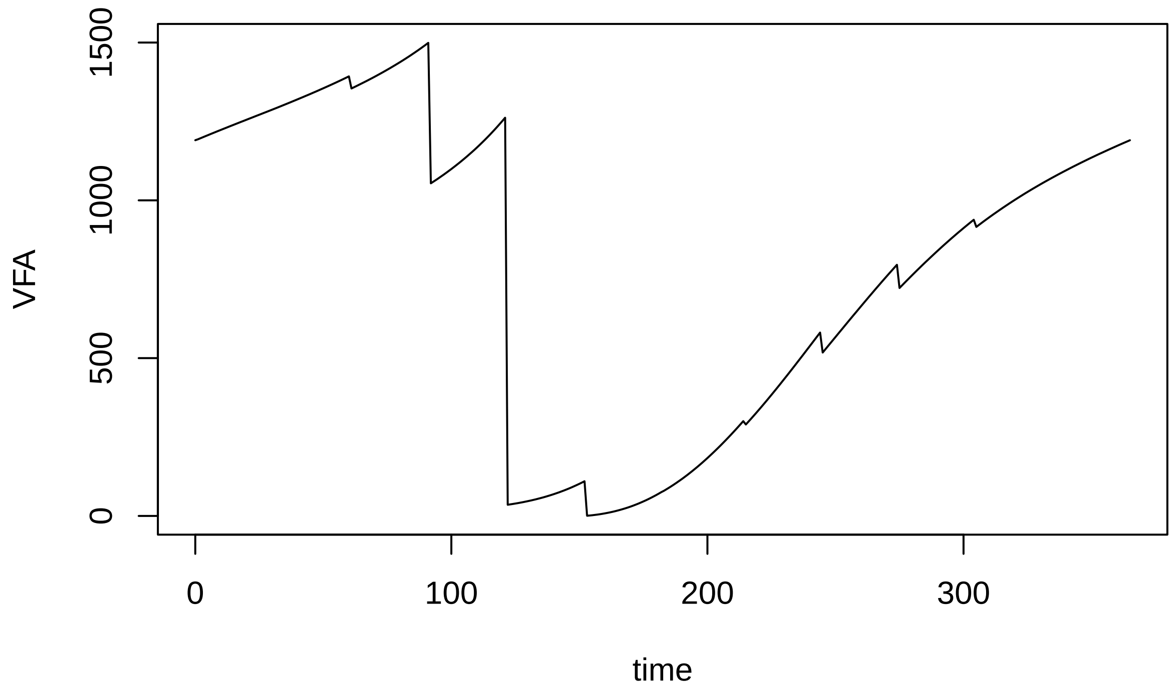

```
matplot(
  abm_out0$time,
  abm_out0[, nn <- paste0(grp_pars$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



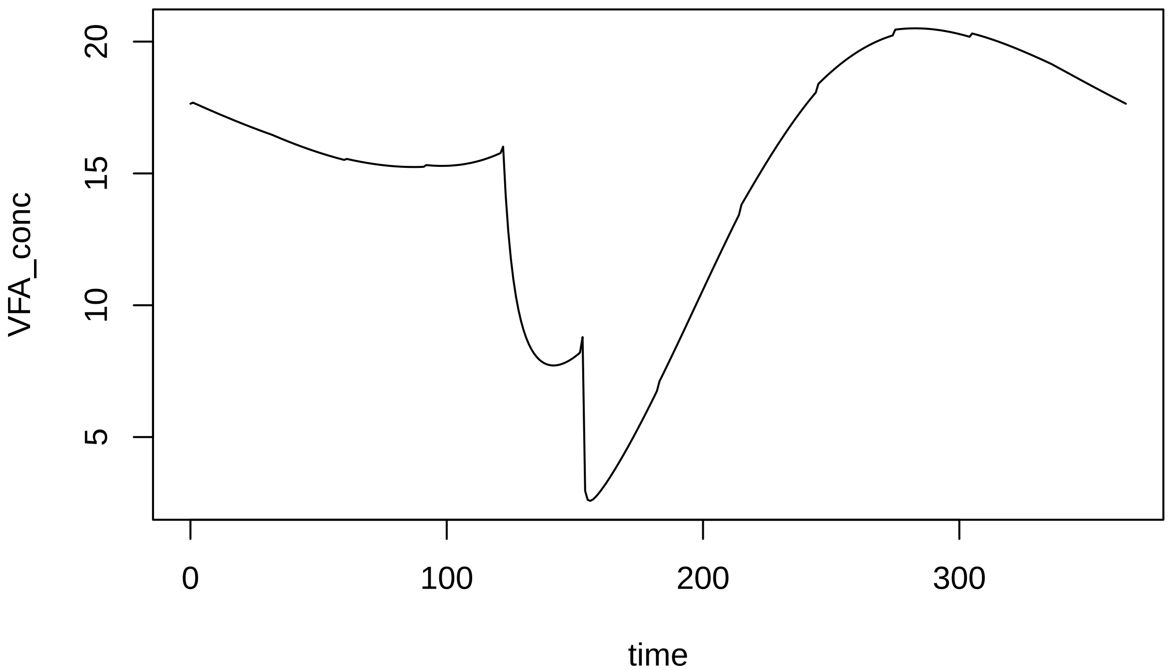

```r
matplot(
  abm_out0$time,
  abm_out0[, nn <- paste0(c('starch', 'Cfat', 'CPs', 'CPf', 'RFd'), '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```




```
plot(VFA ~ time, data = abm_out0, type = 'l')
```




```
plot(VFA_conc ~ time, data = abm_out0, type = 'l')
```


VFA_conc
20
15
10
5
0
100
200
300
time